

TAHA BIN YOUSUF

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Summary

Full-Stack Systems Engineer, AI Developer, and Co-Founder at **SKETCH2STACK** with proven experience architecting high-throughput point-of-sale platforms, telematics microservices, and distributed web applications. Strong engineering background spanning Python (FastAPI, Django), modern JavaScript/TypeScript (Next.js App Router, React.js), C++ with OpenMP parallel computing, and normalized relational database modeling (PostgreSQL, SQL). Experienced in building offline-resilient local sync engines, ESC/POS hardware integrations, automated reporting pipelines, and production machine learning models.

Education

COMSATS University Islamabad, Lahore Campus

Bachelor of Science in Software Engineering (BS SE) — Class of 2027

Lahore, Pakistan

2023 – Present

- **Core Engineering Coursework:** Data Structures & Algorithms, Operating Systems, Database Systems (3NF/BCNF), Object-Oriented Software Engineering, Software Architecture & Design, Machine Learning, Parallel & Distributed Computing, Web Engineering, Mobile App Development.

Work Experience

SKETCH2STACK

Co-Founder & Systems Lead

Lahore, Pakistan

Jun 2026 – Present

- Co-founded an independent software studio delivering 5+ production-grade web, mobile, and backend systems across wholesale retail, food service, luxury jewelry eCommerce, commercial real estate, and automotive industries.
- Engineered full-stack architectures using Next.js (App Router), FastAPI, Node.js, and PostgreSQL; designed normalized schemas, transactional integrity ledgers, background worker queues, and automated CI/CD deployment pipelines.
- Implemented offline-resilient local sync mechanisms using SQLite and indexed client stores, ESC/POS thermal receipt printing pipelines, hardware barcode scanner listeners (<50ms), and granular Role-Based Access Control (RBAC).

Solution A & I

Full Stack Python Engineer Intern

Remote

Aug 2026 – Oct 2026

- Executed Django REST API migration from a legacy Laravel fleet-management platform; developed core business logic for logistics dispatches, multi-stop deliveries, GPS device management, and secure JWT authentication sessions.
- Engineered a fleet telematics analytics and reporting engine featuring 18+ endpoints: Plant-to-Warehouse (PTW) supply-chain tracking, geofence compliance calculators, driver safety metrics, POI coordinate mapping, and ReportLab PDF export generators.
- Applied enterprise Leaf-Pattern service layers, wrapped database mutations in atomic transactions (`@transaction.atomic()`), eliminated N+1 query bottlenecks via `.select_related()` and `.prefetch_related()`, and standardized external Traccar GPS gateway integrations.

Arch Technologies

Machine Learning Intern

Remote

Aug 2026 – Sep 2026

- Developed, trained, and benchmarked classical machine learning and deep learning pipelines in Python using Scikit-Learn, Pandas, NumPy, and TensorFlow.
- Constructed an end-to-end NLP spam detection pipeline (TF-IDF vectorization with Naive Bayes), a Convolutional Neural Network (CNN) on MNIST handwritten digits achieving 98.4% test accuracy, and California Housing price regression models.

Fentix Tech

Full-Stack Web Development Intern

Remote

Sep 2026 – Oct 2026

- Completed intensive full-stack development track focusing on HTML5, CSS3, Bootstrap, Responsive UI/UX Design, JavaScript (ES6+), React.js, Node.js, and MongoDB.
- Engineered modular, responsive front-end interfaces in React.js and Bootstrap alongside RESTful backend APIs in Node.js and Express with secure JWT authentication middleware, CORS handling, and transactional MongoDB models.

Commercial Software Systems & Production Platforms

Next-Gen Clinic & Pharmacy Management OS (Independent Commercial Venture)

Recent (2026)

Next.js 16, FastAPI, Python, PostgreSQL, PWA, GS1/GTIN 2D QR Parser, Gemini Vision OCR

- **Shipped Production Core:** Engineered a high-throughput Clinic & Pharmacy POS system for clinical practices and retail pharmacy counters, featuring sub-50ms hardware barcode scanner interception, Pakistani 14-digit QR GTIN parsing, and granular Role-Based Access Control (RBAC).
- **Smart Inventory & FEFO Batching:** Implemented strict First-Expired, First-Out (FEFO) batch deduction logic across pharmaceutical stock, completely eliminating expired drug stock liability and automating shelf batch rotation.
- **Multimodal Invoice Extraction:** Integrated Google Gemini Multimodal OCR to extract medicine brand names, dosages, batch numbers, and expiry dates directly from physical supplier invoices and packing slip PDFs into digital accounting ledgers.
- **Scalable Roadmap Architecture:** Designed modular decoupled roadmap deployable as standalone clinical EHR (OPD queues, digital prescriptions), retail pharmacy POS, centralized supplier procurement hub, and live multi-branch geolocation map network.

Bakery Wholesale Management & Dispatch POS (SKETCH2STACK)

2026

Next.js, Node.js, PostgreSQL, Tailwind CSS, ESC/POS Thermal Printing, Offline SQLite

- **Wholesale Transaction Engine:** Built a rapid wholesale billing application designed for loading-dock environments, featuring multi-tier bulk pricing matrices, automated wholesale invoice generation, and customer credit balance ledgers.
- **Hardware Printer Integration:** Engineered direct ESC/POS thermal receipt printer routing with customized template formatting for high-speed counter receipts and loading-bay dispatch slips.
- **Offline-First Architecture:** Designed change-set synchronization with local browser SQLite storage, ensuring cashiers and dispatch managers can execute transactions with zero disruption during network outages.

Pizza Avenue Quick-Service Restaurant POS (SKETCH2STACK)

2026

React.js, Node.js, Express, MongoDB, Tailwind CSS, Real-Time WebSockets

- **High-Volume Order Engine:** Architected a fast-paced restaurant POS handling complex menu customizations, add-on crust modifiers, deal combos, and instant table-side order entry.
- **Kitchen Display System (KDS):** Implemented real-time order status broadcasting via WebSockets to kitchen screens, optimizing order preparation times and eliminating lost paper tickets.
- **Shift Reconciliation & Cash Balancing:** Engineered end-of-shift drawer reconciliation modules with immutable cash balancing, sales category breakdowns, and automated daily manager summaries.

Housing Society Master-Plan Management Portal (SKETCH2STACK)

2026

Next.js 15+ (App Router), React.js, Node.js, MongoDB, Responsive UI/UX

- **Commercial Real Estate Management:** Built a commercial land development portal for Market Sikander Chowk, streamlining master-plan plot records, phase conversion workflows, and buyer allocation ledgers.
- **Dynamic Allocation Registry:** Developed interactive plot status dashboards, search indexing across commercial zones, and verifiable digital allocation receipts.

Luxury Jewelry Boutique E-Commerce Platform (SKETCH2STACK)

2026

Next.js, React.js, Tailwind CSS, RESTful APIs, Cloud Media Storage

- **High-End Digital Showcase:** Developed an elegant luxury jewelry e-commerce showcase featuring high-resolution catalog displays, dynamic product inquiry workflows, and bespoke visual branding.
- **Inquiry & Client Acquisition:** Engineered custom quotation forms, customer inquiry routing, and mobile-responsive catalog exploration tailored for luxury goods.

CarZone Automotive Dealership Showroom (SKETCH2STACK)

2026

React.js, Node.js, Express, MongoDB, Tailwind CSS

- **Automotive Inventory Engine:** Engineered a comprehensive dealership vehicle inventory management system supporting multi-attribute technical spec filtering, high-resolution media galleries, and vehicle status tracking.
- **Lead Management System:** Implemented customer test-drive booking forms, automated lead capture pipelines, and administrative vehicle pricing controls.

CozyNest Mobile E-Commerce Application (Mobile Systems)

2025

Flutter, Dart, Firebase Authentication, Cloud Firestore, Stateful Cart

- **Cross-Platform Mobile Client:** Engineered a cross-platform mobile shopping application in Flutter with secure Firebase Authentication, dynamic product search, and persistent checkout cart.
- **Real-Time Synchronization:** Integrated Cloud Firestore real-time inventory synchronization, instant cart status updates, and secure user session persistence.

Capstone FYP & Systems Engineering Projects

- DIVA AI: AI Bridal Virtual Try-On Platform** (Capstone FYP — Collaborative Team Project) 2026 – Present
FastAPI, Python, PyTorch, Computer Vision, Conversational AI, Normalized SQL Database
- **Collaborative Final Year Project:** Developing an AI-powered bridal styling and virtual try-on platform for South Asian wedding couture at COMSATS University Islamabad, Lahore Campus.
 - **Conversational Styling Assistant Chatbot:** Personally designed and implemented an intelligent conversational AI chatbot to guide brides through garment discovery, wedding theme curation, and personalized recommendation flows.
 - **Database Architecture & Schema Design:** Architected normalized database models managing multi-vendor bridal designer catalogs, client styling profiles, garment metadata, and measurement records.
 - **Backend API Engineering:** Developed scalable RESTful backend microservices in FastAPI under Agile sprints, interfacing the virtual try-on computer vision pipeline with e-commerce transaction workflows.
- Parallel Linear Regression Matrix Suite** (High-Performance Computing) 2026
C++, OpenMP, Multi-Threading, Cache Optimization, Batch Gradient Descent
- **Multithreaded Compute Kernel:** Implemented high-throughput parallel linear regression with batch gradient descent in modern C++ utilizing OpenMP directives to parallelize matrix operations and cost calculations.
 - **Performance Benchmarking:** Achieved a 3.8x execution speedup over sequential baselines on multi-million element feature matrices; optimized CPU cache line utilization and memory access patterns to prevent false sharing.
- Online Order & Relational Inventory Management System** (Database Systems) 2025
Relational SQL, 3NF/BCNF Schema Design, ACID Transactions, Stored Procedures
- **Normalized Schema Modeling:** Designed and normalized a comprehensive relational schema (3NF/BCNF) with strict PK/FK integrity constraints across customers, categories, products, multi-tier orders, and payment ledgers.
 - **Transactional Reliability:** Built ACID-compliant transaction blocks, complex SQL aggregation joins, and indexing strategies preventing inventory overselling during concurrent checkout attempts.
- Movie Recommendation & Streaming Hub** (Web Engineering) 2025
MERN Stack (MongoDB, Express, React, Node.js), Stripe, JazzCash, EasyPaisa
- **Full-Stack Media Portal:** Developed a recommendation web portal featuring personalized genre filtering, user watchlists, and MongoDB aggregation pipelines.
 - **Localized Payment Gateways:** Integrated dual payment infrastructure supporting international card checkout via Stripe alongside localized mobile wallet payments (JazzCash and EasyPaisa).
- Cosmetic Shop Management System** (Object-Oriented Software Engineering) 2024
Java, Swing/JavaFX, OOP Principles, Relational Data Schemas, File Ledgers
- **Desktop Enterprise Application:** Built a desktop management GUI applying the four pillars of Object-Oriented Programming (encapsulation, inheritance, polymorphism, and abstraction) for product tracking, supplier ledgers, and cashier sales.
 - **Inventory & Ledger Auditing:** Implemented structured file ledger persistence, supplier purchase order tracking, low-stock alerts, and daily sales auditing.

Technical Skills

Languages: Python, C/C++, JavaScript (ES6+), TypeScript, SQL, Dart, Java, HTML5, CSS3, Bash

Frontend & Mobile: Next.js (App Router), React.js, Flutter, Tailwind CSS, Bootstrap, Redux Toolkit, Responsive UI/UX

Backend & Microservices: FastAPI, Django, Django REST Framework, Node.js, Express.js, RESTful APIs, JWT Auth, Traccar Telematics

AI, Computer Vision & ML: PyTorch, TensorFlow, OpenCV, Scikit-Learn, Pandas, NumPy, Gemini Multimodal API, Prompt Engineering

Databases & Storage: PostgreSQL, SQLite, MongoDB, Firebase Firestore, Mongoose, Relational SQL (3NF/BCNF), Redis

Developer Tools & DevOps: Git, GitHub, Docker, Postman, Linux (Ubuntu), Vercel CI/CD, OpenMP, PWA, Swagger OpenAPI, ESC/POS Hardware

Certifications & Honors

Google — Crash Course on Python — Verified Coursera Credential (Grade Achieved: 91.50%) in core Python, OOP, and automation

Google — Google Prompting Essentials Specialization — 4-Course Verified Specialization in Workflow Automation & AI Prompting

Academic Merit & Coursework Honors — COMSATS University Islamabad, Lahore Campus (Software Engineering Department)